

Lecture 1.5 – Types of Data in R

Specific Learning Objectives:

- 1.1.3 – Understand the basic syntax of the R language.
- 1.1.5 – Learn how to assign variables in R's environment.
- 1.1.9 – Understand the data classes of R.

How can you use GenAI to help you learn R?

- **Get advice on how to structure your data or what data types to use for a problem:**
 - Open ChatGPT and ask it: “I am doing a statistic in R and I need to store words and numbers. What would be the best data types to use?”
- **Help troubleshoot specific bits of code that aren't working:**
 - Open ChatGPT and ask it: “I am getting an error with this line of R code, can you explain to me what the error means and how to fix it? The code is: `"y" + 3`”
- **Help study for the skill check by reviewing material:**
 - Open ChatGPT and ask it: “Can you quiz me on the atomic data types of R?”

Check Your Understanding!

Write out an example of each of the five atomic data classes!

Check Your Understanding!

Will the following line of code work without an error?

```
mean(c(T, F, F, F, T, T, T, F))
```

Correct answer

a) Yes!

a) No!

Write about why this is true.

Check Your Understanding!

Assign the following value to a:

```
a <- "y"
```

What class is object a and how can you test this to make sure?

Correct answer

a) numeric
`is.numeric(a)`

c) character
`is.character(a)`

b) logical
`is.logical(a)`

d) complex
`is.complex(a)`

What is R's output if you tried a, b, or d?

In-class exercises

1. In the book, Davies describes logicals as also numbers. Explain how this is true in R.
2. How do you create a string in R? What data type are strings?
3. What is an escape sequence and how are they used in strings?
4. You have the following vector of numbers named `blippi`:

```
blippi <- c(8, 7, 9, 10, 8, 4, 8, 1)
```

Write a line of code that would extract all the numbers that aren't 8. Explain how your code works.

5. Why is it important for R to store information with different data types?

Action Items

- 1. Complete assignment 1.5.**
- 2. Read Davies Ch. 2 for next time.**